

Manufacturing Practices

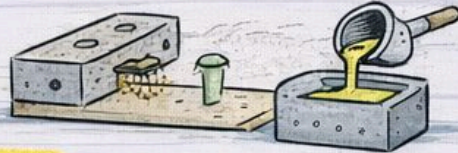
Module-1 Notes

by pyqfort.com

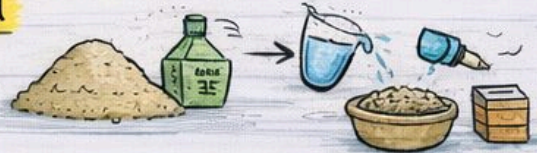


Contents Covered:

- Mould Making Intro

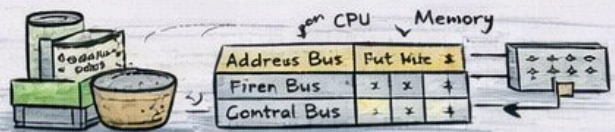


- Constituents of Moulding Sand

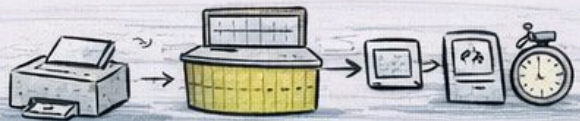


- Sand Preparation

- Properties and Types



- Testing of Moulding Sand

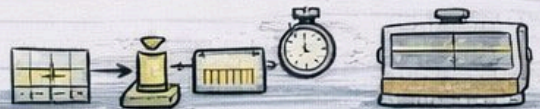


- Types of Moulds

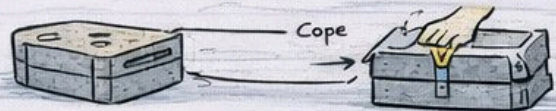
- Moulding Boxes



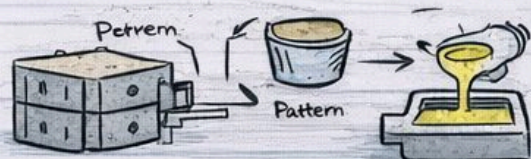
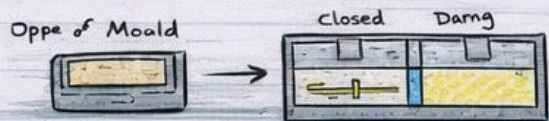
- Moulding Processes



- Moulding Boxes



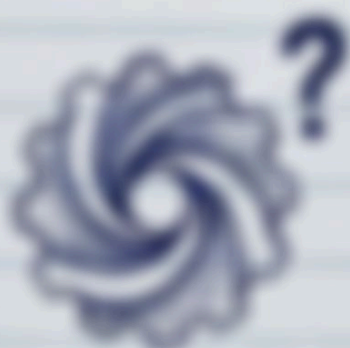
- Moulding Processes



2.1. INTRODUCTION TO CASTING

Casting methods are **universally used** for a **wide variety** of products. **Very small** to **extremely heavy** metal products can be made.

Casting is the **best option** when it is **difficult** or **economically impossible** to produce an item by another process.



DIFFICULT SHAPES



ECONOMICALLY IMPOSSIBLE

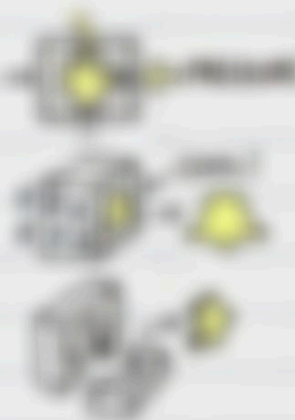
Various **casting processes**

1. **Sand mould** casting ( of sand box).
2. Plaster mould casting.
3. **Metallic mould** casting (Gravity, Slush, Pressure, **Die casting** -  of metal die with molten metal).
4. **Centrifugal casting** 
5. Precision casting.
6. CO₂-mould casting.
7. Continuous casting.

Die Casting Processes

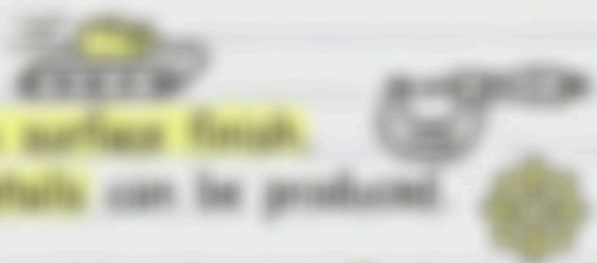
1. The Concept:

- Process of **rapid production of precise castings**.
- **Molten metal under pressure** is forced into **split metal dies** to fill the entire die and details.
- The die is **water cooled**, so casting **solidifies very quickly**.
- The two halves are **separated** and **casting is ejected**.



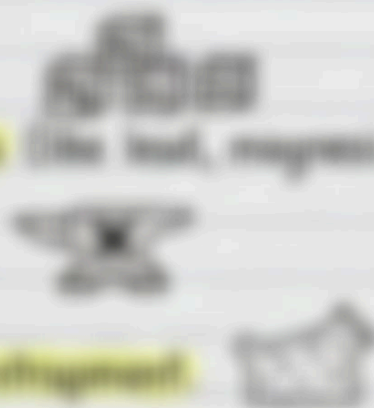
2. Advantages:

- Very **high rate of production**.
- Very **precise dimensions & high surface finish**.
- Very **thin castings with fine details** can be produced.
- Die has very **long life**.
- **Low floor space needed & low cost of casting**.



3. Limitations:

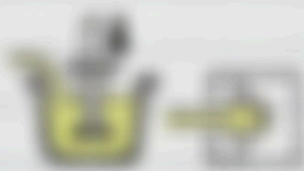
- **Not economical for low batch size**.
- More suitable for **non-ferrous alloys** (like **aluminum, magnesium, tin, zinc**).
- **Heavy castings cannot be produced**.
- **High cost of equipment and die**.
- Castings have **porosity** due to **air entrapment**.



4. Die Casting Machines (Two Types):

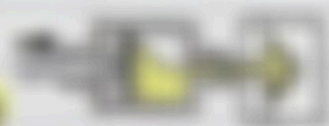
• **Hot Chamber (Plunger Type)**

- Molten metal fills channel by **gravity**.
- **Plunger forces metal** into the die.
- Plunger (made of **refractory material**) operated manually/mechanically/hydraulically.



• **Cold Chamber (Horizontal Plunger)**

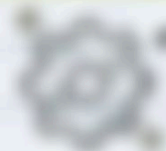
- Chamber operated **pneumatically** or **hydraulically** forces charge into die.
- Suited to **aluminum alloys & high melting temp non-ferrous alloys**.



Precision Casting / Lost Wax Process / Investment Moulding

1. The Concept & Goal:

- Also known as 'lost wax process' or 'investment moulding'.
- Produces very precise castings that do not require subsequent machining.

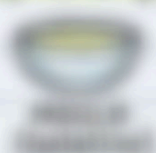
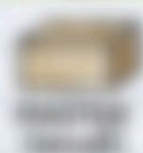


Replaces the Machining method.

2. The Process (Two Stages):

- Stage 1: Preparation of Master Pattern.

- A master pattern (wood/metal) is used to form a mould (positive/negative alloy).
- Liquid wax is poured into this mould to produce the wax pattern ('investment of the pattern').



wax spruing

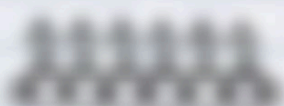
- Stage 2: Wax Pattern & Casting.

- The wax pattern is used to make the final casting mould.
- Crucially, the wax pattern is not taken out. It is melted/lost during the process.



3. Advantages:

- High surface finish, machining eliminated/reduced.
- Very thin castings can be produced.
- Complex shaped castings possible.
- Sound and defect-free.
- Suitable for mass production of small castings.



4. Limitations:

- Not suitable for heavy castings.
- Requires strict control at all stages.
- Very expensive process.



5. Applications:

- Turbines, Motor cars, Sewing machines, Type writers, Calculating machines, Various instruments.



Centrifugal Castings

1. The Concept (Process)

- Molten metal is poured into **rotating moulds**
- Metal is thrown towards the periphery due to **centrifugal force**
- inner surface becomes**

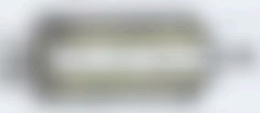


- solidification progresses from the **outer surface inwards**
- All areas of **external/interior** are located at the **inner wall**



2. Three Types

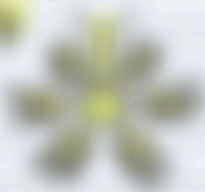
- 1. Fully Centrifugal Casting: Moulds of **uniform geometry** (cylindrical) rotated with **horizontal axis**
- used to produce **long cylindrical castings** (pipes, bushings, gun barrels) **without a central hole**



- 2. Semi-Centrifugal Casting: Moulds rotated about **vertical axis**. Metal fills **upwards** from through gates
- A **downward cone** may be used for the central hole
- used for **large castings** like **gears, disc wheels**



- 3. Centrifugal Castings: Multiple moulds located **radially** about a **central axis**, **horizontal rotation**
- used for **thin-walled** or **small intricate castings**



3. Advantages

- Castings are completely **free from porosity defects**
- long and short** castings
- the structural properties** of properties
- holes, feeders, and cores are **discontinued** (in some types)
- Moulds are **cheap** and **less complicated**
- light** dependent forces to **outer surface** (can be machined out)
- thickness of central wall** can be machined out

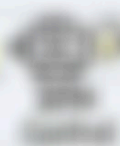
4. Applications

- long cylindrical products** (cast iron pipes for water/sewerage)
- thin gun barrels**
- symmetrical castings** such as **gears, disc wheels, pulleys**

Centrifugal Casting Machines

1. Variable Speed Centrifugal Casting Machine:

- D.C. Motor (3 kW, 400-1200 rpm): Provides **variable speed drive**.



• Key Components:

- **Head stock & Tail stock**: Support the rotating assembly.
- **Clutch plate**: For engaging/disengaging drive.
- **Cooling water spray**: Cools the mould during casting.
- **Holding fixture**: Holds the mould.
- **Feed hopper**: For feeding molten metal.



Sketch as per Fig 1.1

2. Centrifugal Casting Machine with Pneumatic Grips:

- Features **pneumatic grips** for holding the mould.



• Key Components:

- Electric motor & **variable speed drive**: Provide controlled rotation.
- Head stock with **pneumatic gripping mechanism**.
- Chuck, Removable plate, Grip plate, Bush, Connecting rod: Form the **gripping assembly**.
- **Pneumatic cylinder**: Operates the grips.



Sketch as per Fig 1.2

3. Benefits & Speed Selection:

- Ensures **high quality** of casting and **improved productivity**.
- Provides **stepless speed regulation** and **fast mounting** of bearing.
- Speed of bearing rotation (n) is selected by formula:

$$n = \frac{1250}{\sqrt{r}}$$

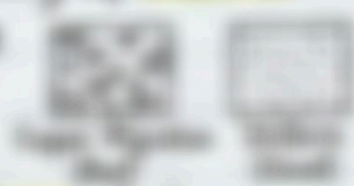
where r = bearing bore radius (mm).



Pressure Casting

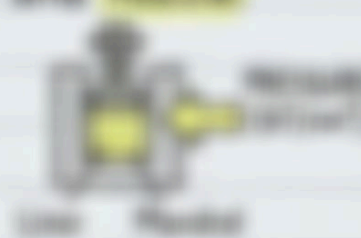
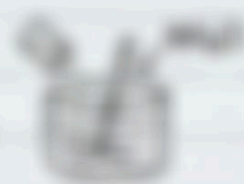
1. Purpose & Goal:

- Used to ensure **high quality** of centrifugal casting of **ballist**.
- Prevents **copper migration** which can occur due to improper rotation speed or cooling rate.
- Achieves a **uniform and homogeneous distribution** of the structural composition of ballist lining.
- Crucial for certain applications requiring high reliability.



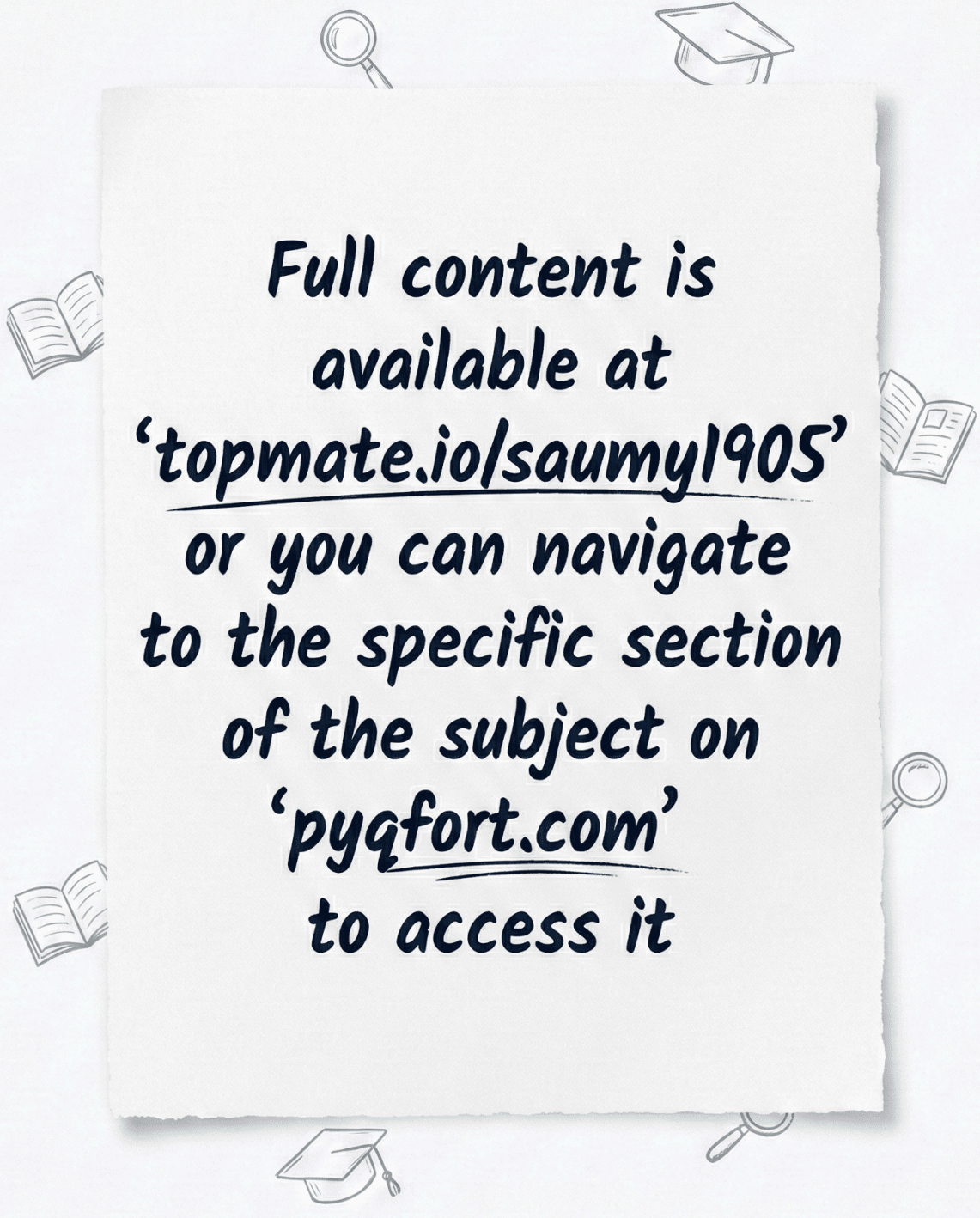
2. The Process (Step-by-Step):

- **Preparation:** Check lining surface of liner. Heat **working plunger** & grip fixture in a portable **electric furnace**. Place assembled liner in **crucible** & **mandrel** (ensuring machining allowance).
- **Pouring & Mixing:** Pour **ballist** and **deoxidizer** by mixing with dry **ammonium chloride** using a propeller.
- **Injection:** Open stop valve.
- **Injection:** Open stop valve. Operate the press. The **working plunger** pushes **melted metal** through a pipeline and **nozzle**.
- **Pressure Application:** Metal is forced into the space between the liner and mandrel at a **high pressure of 6 bar/min**.
- **Cooling & Removal:** Cool liner and grip fixture after a few seconds with **compressed air** and **water spray**. Remove liner for inspection.



3. Key Machine Components:

- **Working plunger & Press:** Apply pressure.
- **Electric crucible & Furnace:** Melt and hold the metal.
- **Pipeline & Nozzle:** Direct the flow.
- **Liner & Mandrel:** Form the casting cavity.



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